SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: INTERNET PROGRAMMING
COURSE CODE: CSA43

COURSE OUTCOME COVERED

CO 3: Implement dynamic web applications by utilizing DOM-based client-side scripting and employing Java Servlets for server-side request processing and response handling. (L3)

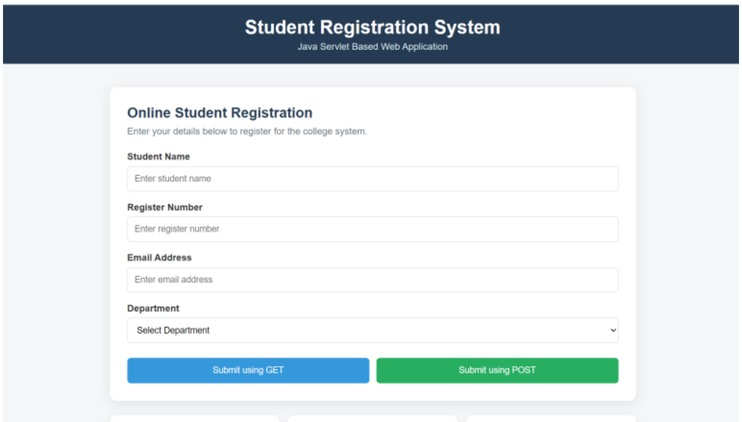
Assessment Tool : Scenario-Based Assessment

Weightage: 40%

QUESTIONS: **Total Marks: 30**

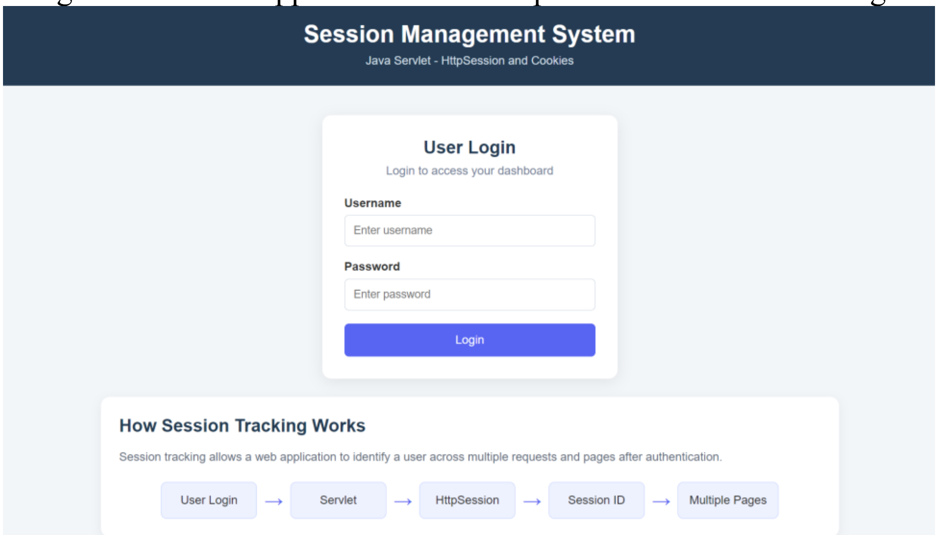
Question 1

Suppose that a college wants to develop an online student registration system where students submit their details through a web form, and the data is processed using a Java Servlet.
Explain the architecture of a servlet-based application. Design and implement a servlet to handle form data using both GET and POST methods. Describe the servlet life cycle (init, service, destroy) and justify which method is more appropriate for handling sensitive data.



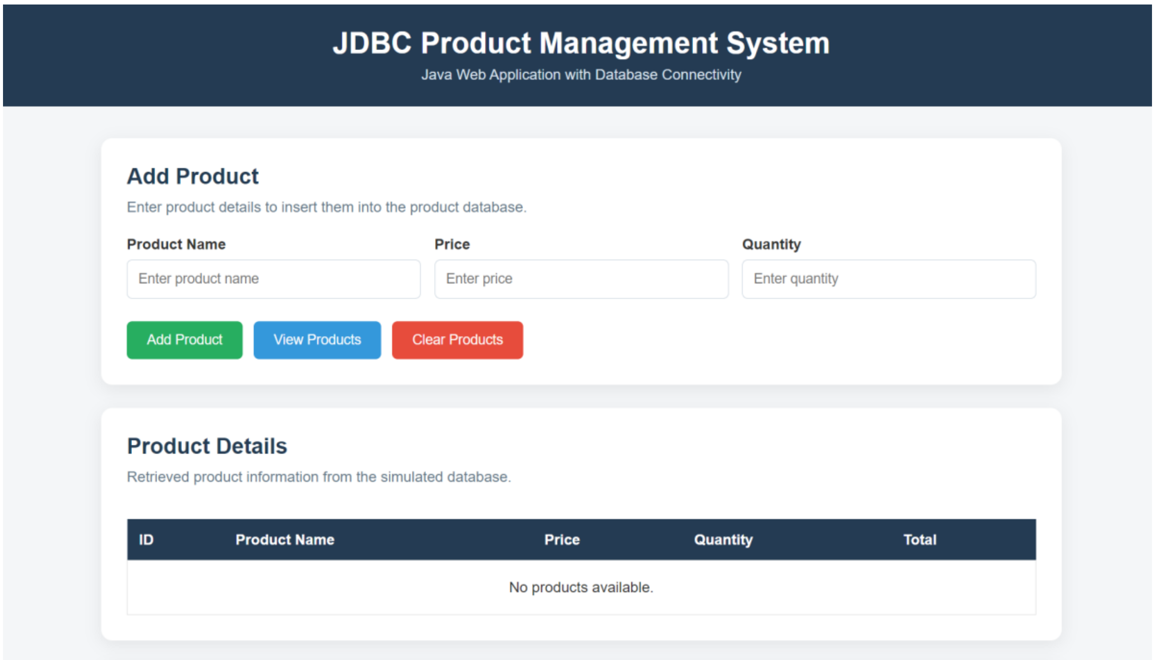
Question 2

Suppose that a website requires users to log in and maintain their session across multiple pages after authentication.
Design a session management system using HttpSession and Cookies in Servlets. Explain how session tracking works in web applications and compare session-based tracking with cookies.



Question 3

Suppose that an e-commerce platform needs to store and retrieve product details from a database using a Java-based web application.
Design a JDBC-based application to connect to a database, insert product data, and retrieve it using SQL queries. Explain the JDBC architecture and describe the steps involved in database connectivity along with proper exception handling.



Code of Conduct:

I **<u>Naveen G / 192411019</u>** *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	**Weightage (%)**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand